Unit: Unit 9: Data Statistical Analysis
Topic: Box Whisker (Dispersion)

Name: ____________

Date: ________

Challenge (Habanero)

Q1. A recipe for cake requires $2 \cdot x$ cups of flour. If x ranges from 1 to 5, what is the interquartile range of flour needed?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q2. The construction of a bridge requires beams of lengths $10, 12, 15, 18, 20$. What is the first quartile of the beam lengths?
(A) 10
(B) 12
(C) 13
(D) 15
(E) 18

Q3. The scores of a cooking competition are $85, 90, 92, 95, 100$. Which score is an outlier?
(A) 85
(B) 90
(C) 92
(D) 95
(E) 100

Q4. A set of measurements for a building are $20, 22, 25, 28, 30$. What is the range of the measurements?
(A) 8
(B) 10
(C) 12
(D) 15
(E) 18

Q5. The time taken to bake a cake at different temperatures is $15, 20, 25, 30, 35$ minutes. What is the third quartile?
(A) 20
(B) 25
(C) 28
(D) 30
(E) 35

Q6. A box-and-whisker plot shows a minimum value of 5 and a maximum value of 15. If the first quartile is 7, what is the interquartile range?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Q7. A dataset of exam scores has a median of 70 and an interquartile range of 20. If a new score of 95 is added, what is the new median?
(A) 70
(B) 72
(C) 75
(D) 78
(E) 80

Q8. The amount of sugar needed for a recipe is $2x + 5$ grams. If x ranges from 2 to 6, what is the range of sugar needed?
(A) 10
(B) 12
(C) 14
(D) 16
(E) 18

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the data set $12, 15, 18, 20, 22, 25, 30$ and identify the outliers.

Q10. A set of measurements for a construction project are $10.2, 11.5, 12.8, 14.1, 15.6$. Calculate the interquartile range and identify the quartiles.

Chef's Tips

- Use a ruler to draw the box-and-whisker plots
- Check for outliers and explain their significance
- Ensure accurate calculations of quartiles and interquartile range